

USE-CASE FLOW SPECIFICATION

Remote Patient Vitals Alert & Monitoring App

UC-04 — Generate Critical Alert

This is the best core use case because the central purpose of your scenario is detecting abnormal vitals and escalating emergency alerts.

Basic information

- Use Case ID: UC-04
- Use Case Name: Generate Critical Alert
- Primary Actor: On-Call Caregiver
- Supporting Actor: Vital Monitoring Device

Preconditions

1. The patient is registered in the system.
2. The patient's vital monitoring device is connected and transmitting telemetry.
3. Clinical thresholds for the patient's vital signs are configured.
4. An On-Call Caregiver is assigned to the patient.

Postconditions

1. The abnormal vital sign is recorded in the system.
2. A critical alert is generated with the patient's information, abnormal vital value, and timestamp.
3. The alert is sent to the assigned On-Call Caregiver.
4. The alert is either acknowledged or escalated if it remains unacknowledged.

Main Success Scenario

1. The Vital Monitoring Device continuously sends the patient's SpO₂, Heart Rate, and Blood Pressure readings to the system.
2. The system receives and records the latest vital measurements.
3. The system evaluates the received vital measurements against the configured clinical thresholds.
4. The system detects that a vital sign has crossed its critical threshold.
5. The system records the abnormal vital value and timestamp.
6. The system generates a critical alert containing the patient identifier, abnormal vital, measured value, and time of detection.
7. The system sends the critical alert to the assigned On-Call Caregiver.
8. The On-Call Caregiver receives the alert and views the abnormal vital information.
9. The caregiver acknowledges the alert.
10. The system records the acknowledgement and marks the alert as acknowledged.
11. The use case ends successfully.

